Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/28/2021
PI Name:	Motil, Brian J. Ph.D.		
Project Title:	Analysis of Packed Bed Reactor-2 Experimental Data from International Space Station		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	06/25/2020	**End Date:**	12/24/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balakotaiah, Vemuri Ph.D. (University of Houston grant 80NSSC20K0830)		
Grant/Contract No.:	Internal Project ; CoI grant 80NSSC20K0830		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

NOTE: Continuation of "Modeling and Analysis of Packed Bed Reactor Experimental Data from International Space Station" Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant 80NSSC18K0813 (formerly "Modeling and Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity" PI Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX14AD28G (and previously NNX10AL37G "Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity"), PI=Dr. Brian Motil; CoI=Dr. Vemuri Balakotaiah. Also included information for project “GAS-LIQUID TWO-PHASE FLOW THROUGH PACKED BEDS IN MICROGRAVITY,” internal project--Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX17AF91G.

A basic understanding of two-phase flows through porous media is of interest in many chemical and biological processing systems as well as many geophysical applications. NASA technologies include adiabatic two-phase flows or two-phase flows with heat transfer that are relevant to life support systems, fuel cells, in-situ resource utilization, heat pipes, materials processing, production of pharmaceutical grade water, and transport of nutrients in soils. A very important unit operation that involves gas-liquid flow through porous media is the fixed Packed Bed Reactor (PBR). Examples of current space systems in which a fixed PBR is used include the Volatile Removal Assembly (VRA), the Integrated Advanced Water Recovery System (AWRS), and IntraVenous Water GENeration system (IVGEN). NASA is currently operating systems with packed bed reactors for ground and flight-based testing with an incomplete understanding of how the reduced gravity environment affects the performance and reliability of these reactors. This is especially critical when gas-liquid flows are involved. The expected outcome of this research effort is to develop a set of guidelines and tools to enable engineers to reliably design and operate packed bed reactors for microgravity as well as the lunar and Martian environments.

Existing approaches for predicting the flow regimes and pressure drop for gas-liquid flow through a PBR are valid only for 1 g, and are based on liquid (or gas) holdup within the bed. Initial

microgravity tests conducted in the reduced gravity aircraft and drop tower clearly show that the liquid holdup is significantly influenced by gravity. Furthermore, most of these models have only been verified using the "trickle flow" regime which does not exist in microgravity. Basic hydrodynamic models developed specifically for reduced gravity have been partially validated within the constraints of aircraft and ground facilities. However, typical flow ranges used for real systems require up to several minutes to develop fully which is far beyond the time scales available for ground tests. In addition, instabilities and hysteresis effects that may exist require long testing durations to establish new operating conditions. Thus, testing on the International Space Station (ISS) is required for complete validation of the semi-empirical models.

Flight experiments can provide critical hydrodynamic information for a project with a broader scope that includes reduced gravity aircraft and ground-based (1 g) experiments. The main objective of the overall project is to develop and validate macroscopic equations that can be used in partial and microgravity conditions to accurately predict flow pattern transitions; pressure drops; and mass transfer rates in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. The hydrodynamic investigations addressed in the Packed Bed Reactor Experiment (PBRE) focus on the transitions between flow regimes (e.g., bubbly-to-pulse flow transition) and the associated pressure gradients over the range of relevant operating regimes of the PBRs (e.g., liquid and gas flow rates, particle sizes, and fluid viscosities). These design tools provide a fundamental understanding of these phenomena in low gravity.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The PBRE provides a fundamental investigation of two phase flow through porous media in the absence of gravity. This enables a direct study of the other forces that impact the flow and reaction rates such as surface tension, inertia, and viscous forces. These new insights will impact reactor designs on Earth.

Task Progress:

(Ed. note April 2021: compiled from CoI Dr. Balakotaiah's 80NSSC20K0830 progress report dated 4/26/2021)
This is an 18-month grant for analyzing the data obtained in the Packed-Bed Reactor Experiment-2 (PBRE-2). The main objective of the Packed Bed Reactor Experiment (PBRE) International Space Station (ISS) project was to better understand the hydrodynamics of two phase flows in porous media in flow regimes dominated by inertia and viscous effects and understanding the role of the interfacial forces in determining the flow dynamics. For this purpose, we developed and validated macroscopic equations that can be used in microgravity conditions to accurately predict flow pattern transitions and pressure drop in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. Single and two-phase flow through porous media is encountered on the ground and in space in numerous applications related to life support systems, fuel cells, chemical/materials processing, and transporting nutrients to plants. These systems operate differently in the microgravity environment encountered in space travel because the density differences no longer cause the phases to separate under the body force of gravity. In the earlier part of this grant, ISS PBRE-1 experimental data was analyzed at University of Houston (UH) to determine the flow pattern transitions in microgravity and compare the results with the same data obtained in aircraft-based experiments. The results of the analysis of the PBRE-1 data were summarized in a recent publication in a refereed journal (See Bibliography section).

The Packed Bed Reactor Experiment-2 (PBRE-2) was developed and flown on the International Space Station (ISS) by NASA to extend a series of fundamental studies of gas-liquid flows through porous media. PBRE-2 extends PBRE-1 results to include a smaller size glass packing (2 mm) to increase the overall pressure drop in the column. The primary reason for this change was to increase the accuracy of the pressure readings for the lower flow rates. Besides, modifications were made to the test facility. During the PBRE-1 testing, small pressure oscillations were observed external to the test bed. To minimize these external disturbances, the inlet mixing head was modified by making the mixing chamber smaller. In addition, several check valves were removed. We have completed a partial analysis of the PBRE-2 data and a brief summary of the results is provided below.

In PBRE-2 we focused on the whole flow range of the gas and liquid flow rates instead of the only low flow rate range (V-C regime) which resulted in more accurate and reliable data. In the absence of gravity, the interfacial or capillary forces play a more significant role in determining the operational variables such as phase distribution, liquid holdup, and pressure drop, especially when the liquid inertia and viscous forces are at minimum. We used dimensionless analysis and proposed a modified model based on the more accurate experimental data obtained in PBRE-2 runs for predicting the two phase friction factor in porous media. In PBRE-2 we also repeated all tests with a high gas flush before each test condition in addition to high liquid flush runs, since liquid-only flush did not show the same effect in 0-g experiments and could not remove all stuck bubbles from the bed. Higher pressure gradients were observed in the liquid-flush runs comparing to the gas-flush runs. It was probably because of higher gas holdups and stuck bubbles in liquid-flush runs that were released from the bed in gas-flush runs and led to lower gas holdups and pressure gradients. The developed correlation also fitted the experimental data over the full range of both gas and liquid flows and the accuracy was high at higher flow rates where liquid flush and gas flush data approach together.

The capillary contribution of the total pressure gradient was also compared between PBRE-1 3 mm sized particles and PBRE-2 2 mm sized particles versus liquid and gas superficial velocities. It was observed that this contribution is almost 2 times higher for the PBRE-2 data which is also consistent with the proposed empirical correlation.

The UH team has also proposed an initial draft for a publication summarizing some of the main findings of the PBRE-2 experiment and is currently updating this draft in anticipation of the upcoming meeting with the NASA team.

Bibliography Type: Description: (Last Updated: 04/26/2024)

Abstracts for Journals and Proceedings

Motil BJ, Nahra H, Taghavi M, Balakotaiah V. "Packed Bed Reactor Experiment (PBRE)-2: Pressure Drop Measurements in Microgravity." 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020.
Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020

Abstracts for Journals and Proceedings

Taghavi M, Balakotaiah V, Motil BJ. "Pressure drop in bubbly gas-liquid flows in packed-beds in zero gravity: Analysis of experimental results from the International Space Station." 2020 Virtual AIChE Annual Meeting, November 16-20, 2020.
2020 Virtual AIChE Annual Meeting proceedings. November 16-20, 2020.
https://www.aiche.org/academy/conferences/aiche-annual-meeting/2020/proceeding/paper/319g-pressure-drop-bubbly-gas-liquid-flows-packed-beds-zero-gravity-analysis-experimental-results ; accessed 4/28/21. , Nov-2020

Articles in Peer-reviewed Journals

Motil BJ, Ramé E, Salgi P, Taghavi M, Balakotaiah V. "Gas-liquid flows through porous media in microgravity: The International Space Station Packed Bed Reactor Experiment." AIChE J. 2021 Jan;67(1):e17031. https://doi.org/10.1002/aic.17031 , Jan-2021

Awards

Motil BJ, Rame E, Salgi P, Taghavi M, Balakotaiah V. "Editor's Choice: Transport Phenomena and Fluid Mechanics, for article, "Gas-liquid flows through porous media in microgravity: The International Space Station Packed Bed Reactor Experiment". AIChE J. 2021 Jan;67(1)::e17031. https://aiche.onlinelibrary.wiley.com/toc/15475905/2021/67/1 " Jan-2021